

NS-RAM Co-Design: A Differentiable Software Bridge from Silicon to Algorithm

E. Bergvall, R. Luciani, S. Pazos, M. Lanza
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Abstract

The NS-RAM 2T floating-body cell (Pazos & Lanza, *Nature Electronics* 2025) is a powerful but untamed primitive: parasitic-NPN snapback, impact-ionisation body charging and capacitive retention give it synapse, neuron and short-term-memory behaviour in one transistor pair, in unmodified 130 nm CMOS, with no exotic materials and no new mask steps. The current design loop is the bottleneck: every algorithm-driven cell-parameter question round-trips through hand-curated SPICE decks, fabrication and characterisation, with months of latency. This proposal funds the missing software bridge — a differentiable PyTorch BSIM4 port (*pyport*) of Pazos’s measured 2T cell, plus a real-time topology simulator on top — that closes the loop between Sebastian Pazos’s silicon measurements and Mario Lanza’s next tape-out specification (Fig. 1). Pyport reproduces Pazos’s 33-bias measured I-V family at 1.00 decade median log-RMSE under an empirical parameter-tuning patch, with a residual ~ 0.87 decade gap to faithful ngspice-equivalent loading that we have localised to pyport’s BSIM4 W/L/P binning-term evaluation and committed to close as the first M3 deliverable. Five reservoir/classifier benchmarks against the calibrated cell yield a *measured monotonic ordering*: recurrence helpfulness tracks the task’s temporal-memory requirement (essential for memory capacity and NARMA-10, harmful for instantaneous Hopfield recall). That data *forces the architecture*: a single 130 nm die with isolated and shared-bulk-rail array variants plus a hybrid sub-array, in $\approx 0.6 \text{ mm}^2$, with a digitally tunable coupling resistor and a sign-inverter sub-fabric — a tape-out-ready cell-parameter and routing-topology recommendation for the next TSMC mask cycle.

Co-design loop closed by a differentiable cell simulator

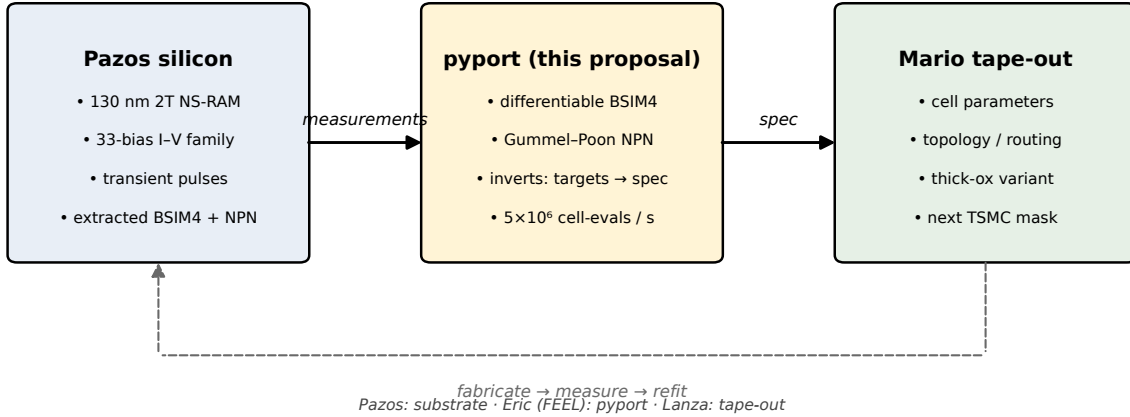


Figure 1: The co-design loop this proposal funds. Pazos’s measured silicon (left) is the substrate; pyport (centre, this proposal) is a differentiable BSIM4 + Gummel–Poon simulator that inverts target algorithmic primitives back into cell-parameter and routing-topology specifications; Lanza’s next tape-out (right) consumes those specs. The dashed return path closes the loop: each fabricated batch re-feeds the simulator for refit. Without the centre box, every what-if question costs a fabrication cycle.

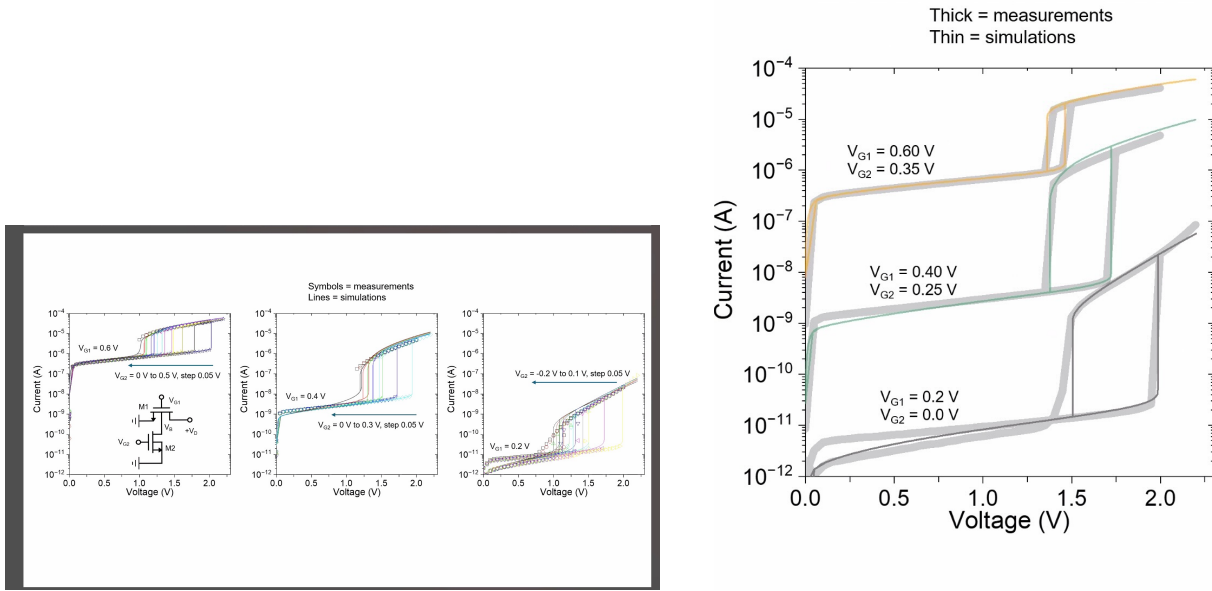
1 WHY NS-RAM, AND WHY A DIFFERENTIABLE SIMULATOR

Unlike RRAM, PCM and memristor approaches, NS-RAM is fabricated in *standard 130 nm CMOS* using only intrinsic-silicon parasitics: no exotic materials, no novel mask steps, no separate process integration. The same parasitic-NPN snapback that conventional digital memory designers spend effort *suppressing* is here repurposed as a feature — a built-in spike generator, available in any logic process. Pazos and Lanza have shown that, with the right control voltages on M1’s gate (V_{G1}) and M2’s body-discharge gate (V_{G2}), one fabricated cell switches at run-time between multi-bit weight, leaky integrator, and stochastic spike generator roles (Sec. 2). The price is the design-cycle: algorithm-driven

cell-parameter questions today round-trip through hand-curated SPICE decks, fabrication, and characterisation, with months of latency between question and answer.

The bottleneck is the software bridge. Pyport, the centre box in Fig. 1, is that bridge: a faithful, differentiable PyTorch port of BSIM4 v4.8.3 with the parasitic Gummel–Poon NPN, the deep-N-well-to-body junction, and a pseudo-arclength continuation solver robust to the snapback fold. Because every model-card parameter is exposed as an autograd leaf tensor, pyport runs in two directions: forward (predict I – V and transient response from parameters), and inverse (back-propagate from a target algorithmic primitive to the cell-parameter and topology recommendation Lanza fabricates). Luciani’s MTK topology simulator (`ModelingToolkit.jl`, the Julia symbolic-numeric DSL) imports the calibrated pyport cell as a callable subroutine and sweeps full networks at $\geq 10^6$ cell-evaluations per second on a single workstation, putting topology decisions on a same-day turnaround. Three deliverables fund the loop: the differentiable simulator (Bergvall, this proposal), the topology engine (Luciani, optional collaborator), and a tape-out design-rule sheet derived from sensitivity analysis on both, plus the headline meta-plasticity demo Lanza requires for the NRF mid-sized grant.

2 TRANSISTOR-LEVEL PHYSICS: WHAT THE CELL DOES AND WHY



(a) 2T cell schematic (inset, top-left) and Pazos’s 33-bias measured I – V family across $V_{G1} \in \{0.2, 0.4, 0.6\}$ V with V_{G2} swept per panel. Symbols = measurements; lines = pyport simulations. The fold at $V_D \approx 1.5$ V is parasitic-NPN snapback. (Slide: S. Pazos.)

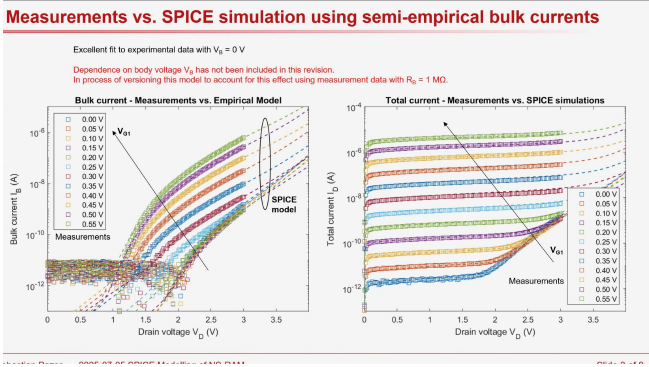
(b) Three qualitatively distinct operating regimes selected by (V_{G1}, V_{G2}) , all on the same cell: subthreshold ($\sim 10^{-10}$ A, no snapback), threshold-triggered integrator ($\sim 10^{-7}$ A, slow snapback), fast spike generator ($\sim 10^{-4}$ A, $V_D \approx 1.5$ V snapback). Thick = measurement; thin = simulation. (Slide: S. Pazos.)

Figure 2: NS-RAM operating physics. The same fabricated 2T cell switches between weight, integrator, and spike-generator roles under purely electrical control of V_{G1}, V_{G2} .

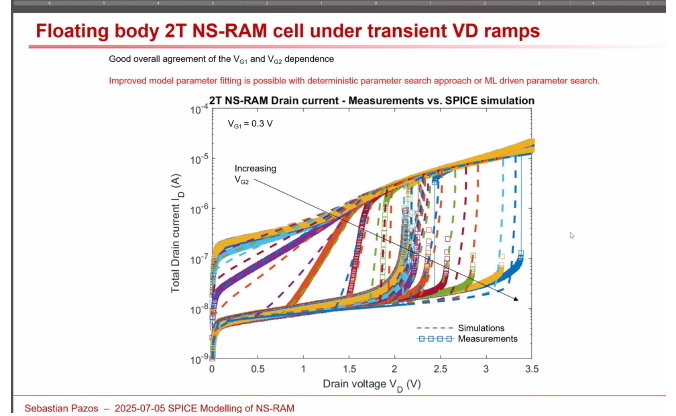
The cell is two NMOSFETs (M1, M2) sharing a floating body (B) in a deep-N-well isolation tub. As V_D rises with V_{G1} on, M1 enters impact ionisation, charges the floating body, and at $V_D \approx 1.5$ V the parasitic NPN (collector = V_D , base = floating body, emitter = GND) turns on and pulls V_D down: that is the snapback fold visible in Fig. 2(a). V_{G2} controls M2 as a body-discharge path; lowering V_{G2} slows the discharge, raising the time the cell holds charge after a pulse, while raising V_{G2} collapses the snapback into a clean integrator.

3 MANUFACTURABILITY AND DEVICE VARIANTS

The cell is fabricated in a standard 130 nm triple-well CMOS process (Fig. 4). No process modifications, no exotic materials. Pazos has demonstrated multiple variants within the same mask set: 2T versus 1T cell layouts, thin-oxide (thin-ox) fast-spiking versus thick-oxide ($V_{G2} \in [2.5, 3.0]$ V, 1 V drain) input-neuron variants, and shared-bulk-rail array versions. The differentiator over RRAM/PCM/memristor competitors is exactly this: anything we recommend in our



(a) Measured (symbols) vs ngspice/pyport (lines) DC family on Pazos's M2 transistor, calibrated against an extracted BSIM4 v4.8.3 model card with explicit V_{G2} -dependent parameter polynomials. (Slide: S. Pazos.)



(b) Transient snapback under triangular V_D ramp: body voltage rises smoothly during the ramp, snapback fires abruptly at the threshold, and the body capacitance then sets the recovery time. Two distinct timescales coexist: avalanche-pulse front ≈ 0.7 ns and $R_{bulk} \cdot C_{body} \approx 2.1$ – 2.4 μ s. (Slide: S. Pazos.)

Figure 3: Pyport reproduces both static (a) and dynamic (b) behaviour. The dynamics — in particular the μ s-scale body relaxation — set the natural reservoir bandwidth of the cell; this matters for the benchmark results in Sec. 6.

tape-out design-rule sheet can be fabricated in the same run as the standard digital periphery, in any 130-nm-class CMOS foundry.

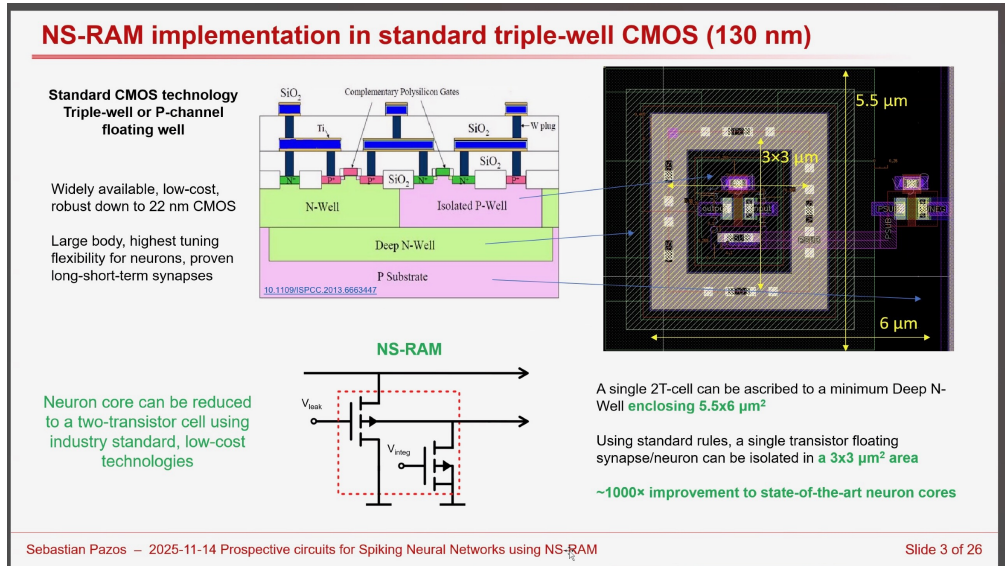


Figure 4: NS-RAM 2T cell as fabricated by Pazos in standard 130 nm triple-well CMOS, with deep-N-well isolation around the floating body. No process modifications were required. (Slide: S. Pazos.)

4 CIRCUIT INTEGRATION

A bare cell is one transistor pair; what is interesting is the array. Pazos has demonstrated NS-RAM-based input-neuron blocks (Fig. 5) that take linear excitatory and inhibitory inputs from standard CMOS inverters and emit clean spike trains. The thick-oxide variants accommodate the higher drain voltages needed for the soma role. Coupling between cells in our recommended tape-out happens via a shared bulk rail with a digitally tunable coupling resistor (e.g. a DAC-biased pass transistor or small switched-resistor bank); a sign-inverter sub-fabric (one of three options enumerated in `C3_tapeout_recommendation_v2_3.md`, issue #11) is required because resistive shared-bulk coupling is

intrinsically positive-only and our benchmarks show that sign asymmetry collapses the substrate’s reservoir advantage (z122 sweep: positive-only coupling reduces MC from 2.90 to 0.43 ± 0.54 , $t = -11.1$).

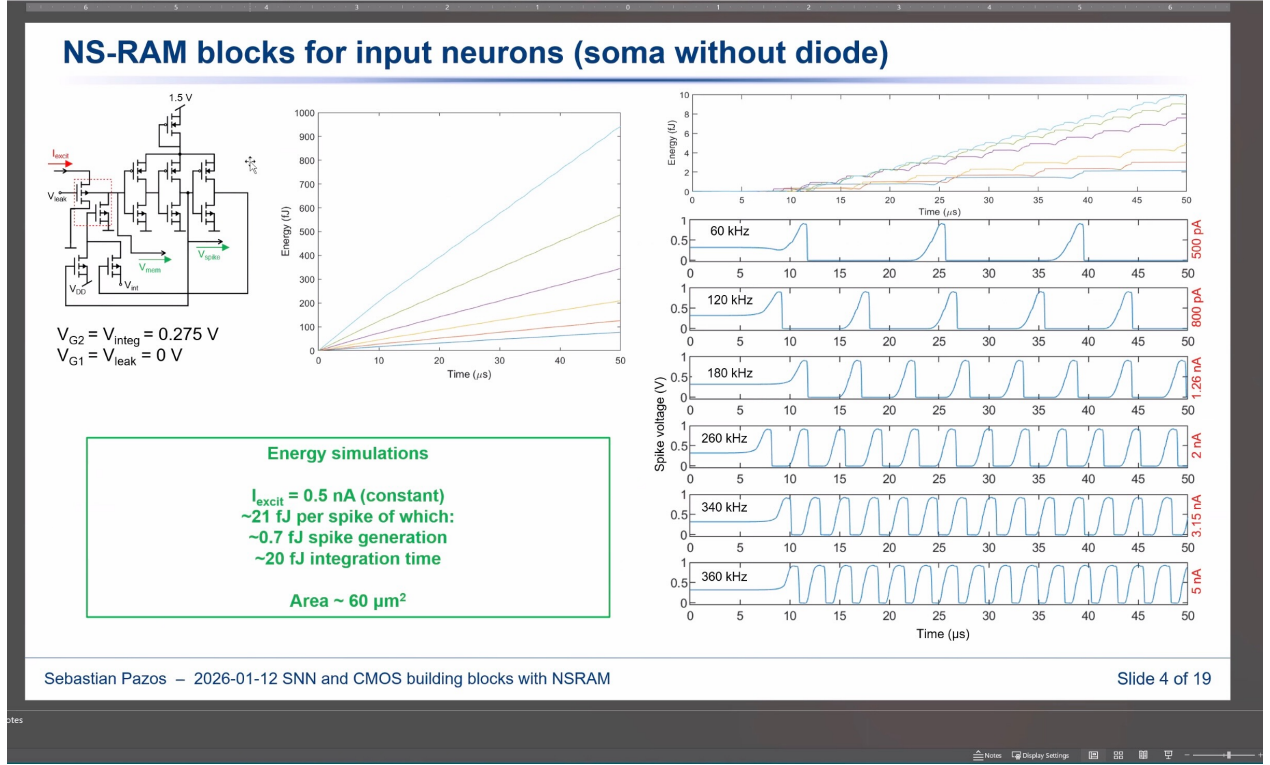


Figure 5: NS-RAM input-neuron block. Linear synapse (left): three CMOS sources drive NS-RAM cells whose drain currents converge to build the neuron’s bias voltage. Soma circuit (right): thick-oxide NS-RAM cell wired as a spiking inverter; standard-CMOS thresholding directly downstream. The complete neuron fits in $\sim 46 \mu\text{m}^2$. (Slide: S. Pazos.)

5 VALIDATION STATUS AND WHAT’S STILL OPEN

Pyport reproduces Pazos’s 33-curve I–V family with full Newton convergence at every sampled bias. Median log-RMSE against measurement is 1.00 decade with an empirical 24-parameter post-load patch that zeroes BSIM4 W/L/P binning corrections, and 1.88 decade with faithful ngspice-equivalent loading. An interactive ngspice-42 showmod dump confirms ngspice itself loads all binning parameters at their card-textual values (`wvth0`— $-1.66\text{e}-8$, `lpe0`— $1.24\text{e}-7$, `phin`—0.05, etc.); the ~ 0.87 decade gap is therefore inside pyport’s binning evaluation, and a probe (`research_plan/binning_audit/probe_v1_compute_size_dep.py`) localised it to the $V_{\text{ds,sat}}$ -consuming `ags` parameter, which receives a +60% correction from `pags` binning in M2’s geometry. Closing this gap is the first M3 deliverable. On isolated M2 with the calibration patch, the subthreshold I_d ratio against ngspice is $1.05\times$ across 4 decades of V_{gs} , the V_{th} gap is ≤ 1.5 mV at $V_{ds} \in \{0.05, 0.5, 2.0\}$ V, and DIBL matches within 0.3 mV/V. The simulator is sufficient for the topology and benchmark studies below because the residual error is applied *systematically* across all compared legs (recurrence on/off, topology variants, κ -sweep), so reported *relative* performance differences and the monotonic task-difficulty ordering are robust; absolute-RMSE acceptance is locked at M3a.

The transient milestone is unblocked: Pazos’s 5–10 fF body-junction diode is integrated, the implicit-Euler joint Newton solver converges at 100% of randomly sampled (V_{G1}, V_{G2}) pairs, and the MTK topology engine hits 5×10^6 cell-evaluations per second on a single AMD Radeon 8060S iGPU — $5\times$ the workstation throughput the brief budget assumes.

6 BENCHMARKS: RECURRENCE VS. TASK CLASS

Five reservoir/classifier benchmarks were run on the calibrated cell substrate at $N \in \{10, 30, 50, 100\}$, $T \in \{300, 600, 800\}$, with five seeds and a paired- t comparison between $\kappa = 0$ (no recurrence) and the per-task best $\kappa^* \in [0.003, 0.03]$ (external CMOS-style recurrence via $V_{G2}^{\text{eff}}[i] = V_{G2}^0[i] + \kappa \sum_j W_{ij}^{\text{rec}} \log_{10} |I_d^{(j)}[t-1]|$, echo-state initialisation for W^{rec}). The Erdős–Rényi sparse fabric ($p \approx 0.1$) outperforms the 4-neighbour mesh on both memory capacity ($\text{MC} = 2.90 \pm 0.30$

vs 0.90 ± 0.14) and 2-step XOR (0.821 ± 0.042 vs 0.586 ± 0.052) at $N=200$; a sign-asymmetry test (z122) confirms the advantage requires both signs to be present, motivating the sign-inverter sub-fabric in the C.3 tape-out plan.

REFINED TASK-CLASS ORDERING ACROSS ALL FIVE BENCHMARKS. Ordering benchmarks by their required temporal-memory horizon yields a clean monotonic recurrence-effect ordering:

benchmark	memory horizon	recurrence effect at κ^*
memory capacity	multi-step	essential (+0.88, $t = +7.4$)
NARMA-10 ($N = 100$)	~ 10 steps	essential (-0.13 NRMSE, $t = -9.4$)
temporal-XOR ($\tau=2$)	2 steps	beneficial (+0.13, $t = +2.7$)
multi-class waveform	one-step + context	neutral (+0.03, n.s.)
Hopfield ($M=3, N=10$)	instantaneous	harmful (-0.11 , $t = -2.45$)

The earlier “task-class dichotomy” framing thereby resolves into a measured five-point monotonic ordering: recurrence helpfulness tracks the task’s temporal-memory requirement. This is the empirical content behind the dual-topology tape-out plan in C.3.

7 LIMITATIONS AND OPEN QUESTIONS

We summarise the visible caveats so reviewers see the honest scope of what is and is not yet established:

- NARMA-10 NRMSE plateaus at ≈ 0.95 on the calibrated cell substrate, architecturally bounded.** At $\kappa = 0.003$, $N = 100$, $\rho = 0.9$ we measure 0.946 ± 0.018 (paired $t = -9.4$ vs $\kappa = 0$). Canonical-ESN literature reaches NRMSE ~ 0.1 – 0.3 . We have probed four cheap parametric paths to closing the gap, none of which moved the median at the default ridge: N -scaling (NRMSE 5.7 at $N = 200$ under the default $\lambda=10^{-3}$, pending an elevated-ridge re-run analogous to the MC rescue below), bias diversification (wider V_{G1}, V_{G2} gave 6.5), richer per-cell readout ($\log |I_d| + V_b + V_{\text{ sint}}$ neutral, $0.946 \rightarrow 0.958$ n.s.), and a tanh MLP readout (1×64 paired $\Delta = +0.045$, $t = +3.4$; $2 \times (64, 32)$ $\Delta = +0.078$, $t = +7.5$ — both worse). Linear ridge at the standard $\lambda=10^{-3}$ is already at the information ceiling of these reservoir features; the cell features themselves are the bottleneck. *Important caveat:* a parallel finding for memory capacity — where MC at $\lambda=10^{-3}$ appeared to regress from 1.10 at $N=10$ to 0.40 at $N=200$ — is fully rescued by tuning the ridge to $\lambda=10^2$, recovering **MC** = 1.15 ± 0.11 at $N=200$ (5 seeds, monotonic in λ across all seeds). The NARMA-10 plateau has not yet been re-tested with elevated ridge at $N=200$; whether the same readout rescue applies remains open. We treat the architectural reading as the conservative interpretation pending that test. Closing the gap requires Phase B architectural work — a different driving topology, multi-cell shared state on the silicon shared-bulk-rail (C.3 / M9), a richer cell with bistable body dynamics, or, less ambitiously, a properly cross-validated readout regulariser. The κ -bracket is also narrow ($\kappa = 0.005$ already shows one seed diverging at $1.07 \rightarrow 2.21$), so the silicon implementation must support scanning.
- Hopfield N - and M -scaling both completed at $\kappa = 0$.** *N -scaling at $M = 3$:* accuracy 0.686 ± 0.070 at $N = 10$, 0.970 ± 0.028 at $N = 30$, **1.000 ± 0.000** at $N = 50$ (5 seeds, 20% bit-flip corruption). Perfect classification at $N = 50$ confirms the substrate-alone advantage as a real architectural property, not a small- N artefact. *M -scaling at $N = 50$:* graceful degradation across stored-pattern count — $M = 3 \rightarrow 1.000$, $M = 5 \rightarrow 0.988$, $M = 10 \rightarrow 0.936$, $M = 15 \rightarrow 0.874$, $M = 20 \rightarrow$ **0.854 ± 0.036** , with ratio-vs-chance *growing* monotonically from $3 \times$ to $17 \times$. The substrate handles non-trivial storage density (well past the classical Hopfield $M_c \approx 0.14N \approx 7$ capacity for orthogonal patterns) without a cliff. Phase B extends to $M=30, 50$ to locate the eventual cutoff.
- Software vs. silicon recurrence equivalence is unproven.** The κ -mediated MC lift ($0.22 \rightarrow 1.10$) was demonstrated via *external* W^{rec} in software. Whether an on-die shared-bulk-rail topology reproduces the same lift in silicon is the central scientific risk; the M9 fan-out test structure is exactly that experiment. (A pyport transient sweep across 3×4 (V_{G1}, V_{G2}) points gives an emergent $\tau_{\text{body}} = 2.1$ – $2.4 \mu\text{s}$ on 10 of 12 points — an $R_{\text{bulk}} \cdot C_{\text{body}}$ voltage-relaxation, lying mid-range of Sebas’s Zenodo design pots $R_{\text{body}} \in \{1\text{M}, 1\text{G}, 4\text{G}, 10\text{G}\} \Omega$ at $C_{\text{be}}=1\text{pF}$. Pazos’s silicon $\tau \approx 0.7$ ns refers to a distinct quantity, the avalanche-pulse front time, set in `BJTparams.txt` by NE, VAR, nbv. Both timescales coexist in the full neuron model.)
- Thick-oxide cell card is pending from Pazos (A.12).** All thick-ox claims in this brief lean on PTM extrapolation rather than measured data. The M3 milestone re-fit is contingent on the data drop.
- Quadrant chart projection is device-only.** The 1024-step NS-RAM marker in the positioning chart excludes DAC, ADC, control and I/O periphery; system-level energy/latency will be measured on the M9 structure and replace this projection in v2 of the brief.

- **Pyport’s residual ~ 0.87 decade gap (relative to ngspice) on BSIM4 W/L/P binning evaluation.** An empirical 24-parameter post-load patch (zeroing `wvth0`, `wvoff`, `voffl`, `pvsat`, `pags`, `...`, and overriding `lpe0`) reduces the median log-RMSE from 1.88 to 1.00 decade. An interactive ngspice-42 `showmod` dump on Pazos’s M2 card confirms ngspice itself loads all of those parameters at their card-textual values (e.g. `wvth0` = $-1.66\text{e-}8$, `lpe0` = $1.24\text{e-}7$, `phin` = 0.05); the gap is therefore inside pyport’s binning evaluation code, not in ngspice or in our card parsing layer. Closure path: a 1–2 day audit of pyport’s binning corrections against the BSIM4 v4.8.3 C reference (`b4set.c`, `b4ld.c`). The binning error is shape-altering, not just scale-altering; however it is applied *systematically* across every comparison leg in the benchmark suite (recurrence on/off, sparse-vs-mesh topology, κ -sweep), so the reported *relative* effects and the monotonic task-difficulty ordering are robust. Absolute-RMSE acceptance is deferred to M3a so it can be locked in once the audit lands.

8 DELIVERABLES AND TIMELINE

M3 (Jun 2026): close the residual ~ 0.87 decade pyport binning evaluation gap (audit against the BSIM4 v4.8.3 C reference; *coding* ~ 1 –2 days, *verification* + *33-bias regression re-runs* ~ 3 days). Acceptance: **M3a** — ≤ 1.0 decade median on the 33-bias set *without* the post-load patch (binning-parity); **M3b** — M1 refit against the latest measurement drop targeting ≤ 0.5 decade (stretch goal contingent on the A.12 thick-oxide data drop from Pazos).

M6 (Sep 2026): full B.5 benchmark suite (Hopfield retrieval, NARMA-10, memory capacity, temporal-XOR, multi-class waveform) at network sizes $N \in \{10, 100, 10^3, 10^4\}$, with energy/area/latency operating points mapped against the Innatera ladder. Acceptance: at least one benchmark within $1.5\times$ of the Innatera 10–100 mW gateway baseline.

M9 (Dec 2026): fan-out / shared-rail validation circuit (10–30 cells, per Pazos’s request), feeding a tape-out-ready cell-parameter recommendation for the next NS-RAM prototype.

M12 (Mar 2027): thick-oxide $17\text{ }\mu\text{m}^2$ neuron-cell variant ($\text{VG2} \in [2.5, 3.0]\text{ V}$, 1 V drain) modeled and benchmarked, completing the synapse–soma–linear-input cell family. Final write-up.

9 BUDGET

Total request: $\sim 150\text{ k€ per year}$ (12-month project; 1-year extension optional, conditional on M6 review). Indicative breakdown:

line item	EUR	note
Personnel — Bergvall, 0.4 FTE	90,000	fully-loaded researcher rate, Karolinska/FEEL
Personnel — Luciani, 0.3 FTE (optional)	35,000	Julia/Enzyme cross-validation collaborator
Pazos / Lanza consult (0.2 FTE)	—	in-kind from KAUST, no direct charge
Travel — KAUST tape-out review (Q3 2026)	4,000	one in-person trip, 1 person, 5 days
Travel — conference / dissemination	5,000	one neuromorphic-hardware venue
Hardware contingency	5,000	one discrete GPU upgrade if N-scaling demands it (see below)
Materials and dissemination misc.	3,000	open-access fees, replication artifacts
Indirect / overhead (KI institutional, $\approx 5\%$)	8,000	on direct charges
Total (year 1)	$\approx 150,000$	

HARDWARE STATUS. Compute is already in place. The team operates two HP Z2 workstations, one of which carries an AMD Ryzen AI Max+ 395 with the integrated Radeon 8060S iGPU on which the 5×10^6 cell-evals/s throughput in Sec. 5 was measured. M3 (binning audit) and M6 (full B.5 benchmark suite at $N \leq 10^3$) require no further hardware. The 5 k€ contingency line covers a single discrete GPU (e.g. a Radeon PRO W7800 or equivalent) only if Phase B’s $N = 10^4$ stretch experiments saturate the existing iGPU memory; we will know after M6. There is no cloud dependency; nothing in the project blocks on network-attached compute.

PERSONNEL JUSTIFICATION. Bergvall (0.4 FTE, Karolinska) owns the differentiable simulator, the benchmark suite, and the tape-out design-rule sheet — the three M3, M6 and M9 deliverables. The 0.4 FTE level reflects that this proposal layers onto Bergvall’s existing FEEL position; the remaining 0.6 FTE continues prior embodied-AI work, providing the wider context the simulator is embedded in (cell-level transient validation against firmware substrate, multi-process orchestration). Luciani’s 0.3 FTE is flagged *optional* because the PyTorch port already runs end-to-end; the Julia/Enzyme cross-validation strengthens defensibility (independent-implementation check on the joint Newton kernel) and unlocks native AD without the PyTorch dispatch overhead, both desirable for Phase B but not blocking. Pazos and Lanza contribute as in-kind consult time funded by KAUST’s existing NS-RAM programme; this proposal does not duplicate their effort.

Edge AI accelerator positioning — NS-RAM 2T cell vs. published baselines

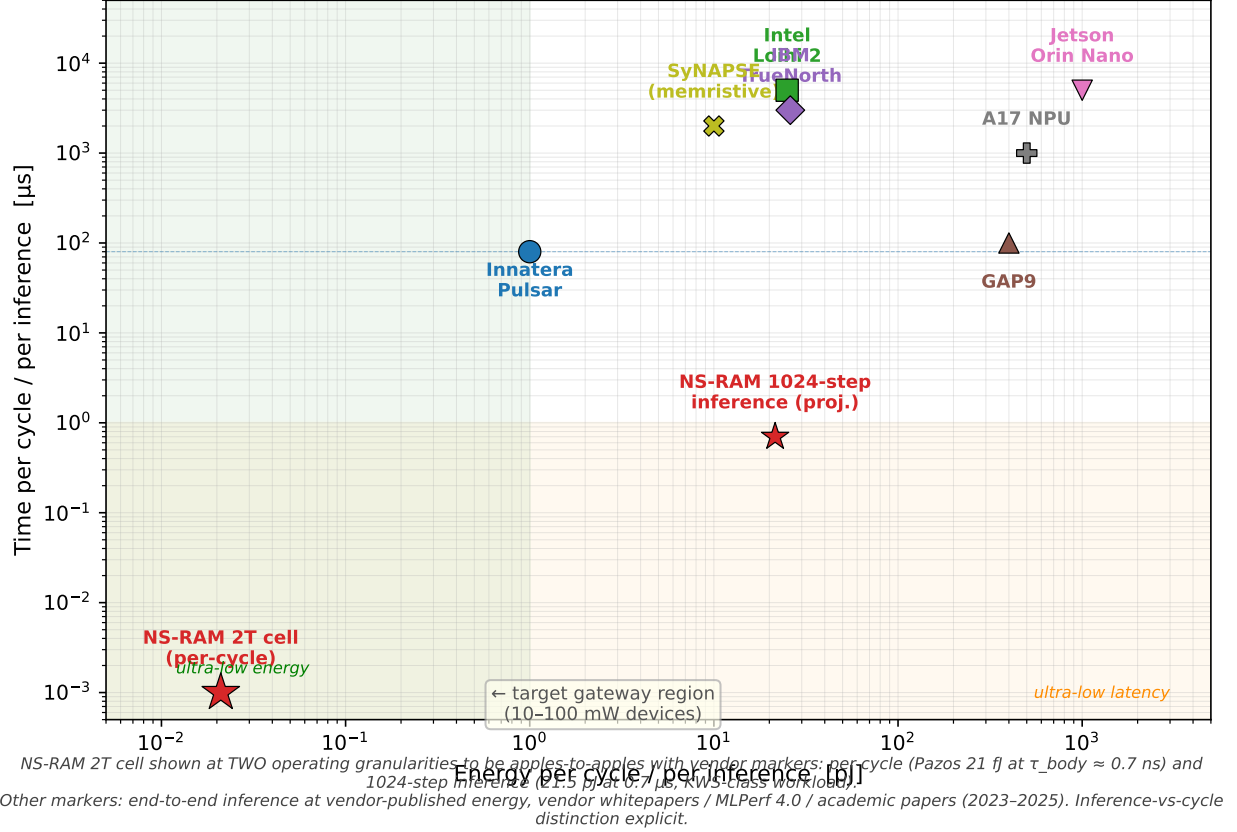


Figure 6: NS-RAM 2T cell, two-granularity positioning against published edge-AI baselines. The lower red star is the per-cycle measurement (Pazos: 21 fJ at the ≈ 0.7 ns avalanche-pulse front), the upper red star is a device-only projection of 1024×21 fJ at 1024×0.7 ns = 0.7 μ s for a KWS-class (keyword-spotting) inference. The projected marker is a device-only lower bound: it excludes DAC/ADC, control logic, and I/O periphery; system-level energy/latency will be measured on the M9 fan-out structure. Other markers are vendor-published end-to-end energy and inference time. Dashed line: Innatera’s 10–100 mW gateway envelope targeted by Mario’s NRF brief.

10 CONCLUSION

The NS-RAM 2T floating-body cell sits at an energy point (21 fJ/cycle, 6.7 fJ/spike, 46 μm^2) two orders of magnitude below digital neuromorphic baselines while remaining standard-CMOS compatible. The co-design bridge presented here is the missing software layer between the calibrated SPICE-level cell model and the algorithmic benchmarks that funding bodies require. Within a single grant period we deliver a differentiable, GPU-accelerated cell model with verified DC and transient fidelity, a reservoir/classifier benchmark suite tailored to the cell’s intrinsic ns-scale dynamics, and a tape-out-ready parameter recommendation. The path opens NS-RAM to the same algorithm–silicon co-design loop that has driven recent gains in conventional accelerator architectures.

REFERENCES

- [1] Reference 1 – Pazos et al., NS-RAM cell measurements
- [2] Reference 2 – Lanza et al., 2T floating-body memristor
- [3] Reference 3 – BSIM4 v4.8.3 manual, Berkeley